

Thiamine-responsive megaloblastic anemia syndrome

Description

Thiamine-responsive megaloblastic anemia syndrome (TRMA) is a rare condition that is characterized by hearing loss, diabetes, and a blood disorder called megaloblastic anemia. Megaloblastic anemia occurs when a person has a low number of red blood cells (anemia), and the red blood cells that are present are larger than normal (megaloblastic). The symptoms of this blood disorder may include decreased appetite, lack of energy, headaches, pale skin, diarrhea, and tingling or numbness in the hands and feet. Individuals with TRMA typically develop megaloblastic anemia between infancy and adolescence. TRMA is called "thiamine-responsive" because the anemia can be treated with daily doses of vitamin B1 (thiamine).

In people with TRMA, hearing loss typically develops during early childhood and is caused by abnormalities of the inner ear (sensorineural hearing loss). The hearing loss usually worsens over time. It remains unclear whether treatment with thiamine can improve hearing or delay hearing loss in people with TRMA.

People with TRMA often develop diabetes mellitus, a condition in which glucose levels can become dangerously high, at an early age. Although some individuals with TRMA develop diabetes during childhood, they do not have the autoimmune form of diabetes that typically develops in children, called type 1 diabetes. People with TRMA usually require insulin to treat their diabetes. In some cases, treatment with thiamine may delay the onset of diabetes or reduce the amount of insulin a person needs.

Some individuals with TRMA have abnormalities of the eye, including optic atrophy, which is the degeneration (atrophy) of the nerves that carry information from the eyes to the brain. Additional features of this condition may include heart and blood vessel (cardiovascular) problems, intellectual disabilities, behavioral changes, and seizures.

Frequency

TRMA is rare, though its exact prevalence is unknown. Approximately 200 individuals with this condition have been described in the medical literature.

Causes

Variants (also called mutations) in the *SLC19A2* gene cause TRMA. This gene provides instructions for making a protein called thiamine transporter 1, which transports thiamine

into cells. Thiamine is found in many different foods and is important for numerous body functions.

Most variants in the *SLC19A2* gene lead to the production of an abnormally short, nonfunctional version of thiamine transporter 1. Other variants change single protein building blocks (amino acids) in thiamine transporter 1. All of these variants disrupt the protein's ability to bring thiamine into the cell. TRMA is treated with daily doses of thiamine because some thiamine can pass into the cell without the help of a transporter when the vitamin is present in large amounts.

It remains unclear exactly how variants in the *SLC19A2* gene cause the specific set of features seen in people with TRMA. Research suggests that an alternative transporter made by a different gene is available to transport thiamine in many tissues of the body. However, this alternative transporter may not be as active in the inner ear, the bone marrow (where blood cells are made), and the pancreas (where insulin is made), which would explain why these tissues are affected in people with TRMA.

[Learn more about the gene associated with Thiamine-responsive megaloblastic anemia syndrome](#)

- *SLC19A2*

Inheritance

This condition is inherited in an autosomal recessive pattern, which means both copies of the gene in each cell must have a variant to cause the disorder. The parents of an individual with an autosomal recessive condition each carry one copy of the altered gene, but they typically do not show signs and symptoms of the condition.

Other Names for This Condition

- Rogers syndrome
- Thiamine-responsive megaloblastic anemia with diabetes mellitus and sensorineural hearing loss
- TRMA

Additional Information & Resources

Genetic Testing Information

- Genetic Testing Registry: Megaloblastic anemia, thiamine-responsive, with diabetes mellitus and sensorineural deafness (<https://www.ncbi.nlm.nih.gov/gtr/conditions/C0342287/>)

[Genetic and Rare Diseases Information Center](#)

- Thiamine-responsive megaloblastic anemia syndrome (<https://rarediseases.info.nih.gov/diseases/9210/index>)

Patient Support and Advocacy Resources

- National Organization for Rare Disorders (NORD) (<https://rarediseases.org/>)

Catalog of Genes and Diseases from OMIM

- THIAMINE-RESPONSIVE MEGALOBlastic ANEMIA SYNDROME; TRMA (<https://omim.org/entry/249270>)

Scientific Articles on PubMed

- PubMed (<https://pubmed.ncbi.nlm.nih.gov/?term=%28Anemia,+Megaloblastic%5BMAJR%5D%29+AND+%28%28thiamine-responsive+megaloblastic+anemia+syndrome%5BTIAB%5D%29+OR+%28thiamine-responsive%5BTIAB%5D+AND+megaloblastic+anemia%5BTIAB%5D+AND+syndrome%5BTIAB%5D%29+OR+%28rogers+syndrome%5BTIAB%5D%29+OR+%28trma%5BTIAB%5D%29+OR+%28thiamine-responsive+myelodysplasia%5BTIAB%5D%29%29+AND+english%5Bla%5D+AND+human%5Bmh%5D+AND+%22last+3600+days%22%5Bdp%5D>)

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